**SC7 Bayes Methods**

**First problem sheet**

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## Section A questions (optional)

1. Let $x_1, x_2, x_3, ...$ be an infinite exchangeable sequence of binary random variables with generative model $q \sim F$ and $x_i \sim p(\cdot|q)$ jointly independent for $i \geq 1$ given $q$. Show that $\text{cov}(x_i, x_j) \geq 0$ for all $i, j \in \{1, 2, 3, ...\}$.

**Solution:** When $i = j$ we have $\text{cov}(x_i, x_i) = \text{var}(x_i) \geq 0$. When $i \neq j$, by exchangeability $E(x_i) = E(x_1)$ for all $i = 1, 2, 3, ...$, and $x_i, x_j$ are independent given $q$ so

$$\begin{aligned}
\text{cov}(x_i, x_j) &= E(x_i x_j) - E(x_i)E(x_j) \\
&= E(E(x_i x_j|q)) - E(x_1)^2 \\
&= E(E(x_i|q)E(x_j|q)) - E(x_1)^2 \\
&= E(E(x_1|q)^2) - E(E(x_1|q))^2 \\
&= \text{var}(E(x_1|q)) \\
&\geq 0.
\end{aligned}$$

2. Let $\Sigma$ be a positive definite $n \times n$ covariance matrix and for $x \in \mathbb{R}^n$ let $p_{1:n}(x) = N(x; 0_n, \Sigma)$. Let $\mathcal{O}_n$ be the set of all non-empty subsets of $[n]$ and let $o \in \mathcal{O}_n$ be a set with $m$ elements. For $x_o = (x_{o_1}, \ldots, x_{o_m})$ with $x_o \in \mathbb{R}^m$ let $p_o(x_o) = N(x_o; 0_m, \Sigma_{o,o})$ where $0_m$ is a vector of $m$ zeros and $\Sigma_{o,o}$ is the sub-matrix of $\Sigma$ obtained by taking the rows and columns in $(o_1, \ldots, o_m)$. Define what it means for the family of distributions $\{p_o; o \in \mathcal{O}_n\}$ to be marginally consistent and show that this holds. *Hint: it might help to take a look at Section 3.1.3 of our notes.*

**Solution:** MC holds if

$$p_o(x_o) = \int_{\mathbb{R}^{n-m}} p_{1:n}(x) dx_{[1:n]\setminus o}$$

for each $o \in \mathcal{O}_n$. Since marginals of multivariate normals $N(\mu, \Sigma)$ are simply obtained by dropping the entries in $\mu$ and $\Sigma$ corresponding to the integrated variables, we have

$$\int_{\mathbb{R}^{n-m}} p_{1:n}(x) dx_{[1:n]\setminus o} = N(x_o; 0_m, \Sigma_{o,o})$$

so the family is MC. To verify this briefly, let $A$ be a $m \times n$ matrix of zeros except that $A_{i,o_i} = 1$ for $i = 1, \ldots, m$ and let $z \sim N(0_n, \Sigma)$. Let $z_o = Az$ so we just extract the variables we want. Linear combinations of normals are normal so we just need the mean and covariance of $z_o$. We have $E(z_o) = 0_m$ and $\text{var}(Az) = A\Sigma A^T = \Sigma_{o,o}$.

## Section B questions

3. In the radiocarbon dating example (with the same key question about span), suppose the dated materials are found in layers (strata) piled up on one another, with $y_{i,j}$ the radiocarbon date for $\theta_{i,j}$, the $j$'th date in the $i$'th layer. Let $L < \psi_1 < \psi_2 < ... < \psi_M < U$ be the age parameters for the layer boundaries. If we have $n_i$ dates from the $i$th layer we know that for $i = 1, 2, ..., M-1$, and $j = 1, 2, ..., n_i$, $\psi_i < \theta_{i,j} < \psi_{i+1}$ (so specimen dates in higher layers are not as old as dates in lower layers). Let $\psi = (\psi_1, ..., \psi_M)$ and $\theta = (\theta_1, ...\theta_{M-1})$ with $\theta_i = (\theta_{i,1}, ..., \theta_{i,n_i})$.

   Derive a prior density $\pi(\theta, \psi)$ for the parameters $\theta, \psi$ with reference to the prior elicitation checklist given in lectures. Hint: how are the layer boundary dates $\psi_2, ..., \psi_{M-1}$ generated?

4. (From Cox and Hinkley *Theoretical Statistics*) For $i = 1, ..., n$, let $\theta_i \in \{0, 1\}$ be the indicator for the event that student $i$ enjoys the course this year and let $\theta = (\theta_1, ..., \theta_n)$. Suppose our prior probability for $\theta_i = 1,\ i = 1, ..., n$ is that they are iid with $P(\theta_i = 1) = p$ with $p$ our prior probability that an individual student enjoys the course and we take a fixed value of $p$ expressing our prior expectation for the proportion enjoying the course (based perhaps on past years).

   Our prior on the function $q(\theta) = n^{-1} \sum_i \theta_i$ has mean $p$ (that's good) and variance $p(1-p)/n$. If $n$ is large this prior expresses near certainty in the proportion enjoying the course. This doesn't represent prior knowledge.

   Modify the prior model so that the variance doesn't go to zero as $n \to \infty$ whilst retaining $E(q(\theta)) = p$. *Hint: your solution will be a hierarchical model of some sort, but beyond that there are many correct answers.*

5. Let $X_1, X_2$ be binary random variables. Table entries below give probabilities, $p(x_1, x_2) = \Pr(X_1 = x_1, X_2 = x_2)$, for outcomes $(X_1, X_2) = (x_1, x_2)$ indicated by row and column.[1]

| | $X_1 = 0$ | $X_1 = 1$ |
|---|---|---|
| $X_2 = 0$ | 0 | 1/2 |
| $X_2 = 1$ | 1/2 | 0 |

   (a) Show that $X_1$ and $X_2$ are exchangeable.

   (b) Show that there does not exist a distribution $F$ such that

$$p(x_1, x_2) = \int_0^1 \prod_{i=1,2} p^{x_i}(1-p)^{1-x_i} dF(p),$$

   ie, de Finetti's theorem need not hold if the exchangeable sequence is finite.

6. Consider two urns. In the first urn there are 50 black balls and 50 red balls. In the second urn there are 100 balls, the number of each color unknown. Suppose the proportion of black balls in the second urn is equal $\phi$.

[1] From P. Diaconis and Freedman D. (1980). *Finite Exchangeable Sequences.* Ann. Probab. v8 p745–764.

Jane's $\phi$-prior, $\pi(\phi)$, satisfies $E(\phi) = 1/2$. Jane is offered a choice of urn and color and *two* balls are drawn (with replacement) from the chosen urn. Jane receives a £1 reward for each ball matching her chosen color. Her utility function is $U(0) = 0, U(1) = v, U(2) = 1$ with $1/2 < v < 1$.

Jane is offered red from the first urn or black from the second.

(a) Show that the expected utility of choosing the second urn given $\phi$ is

$$E(U|\phi) = 2\phi(1-\phi)v + \phi^2.$$

(b) Jane chooses the first urn. Show that this choice maximises the expected utility.

(c) Jane is now offered black from the first urn or red from the second. Show that Jane should again choose the first urn.

## Section C questions (optional)

7. An 18×24 square-foot area of grass in a field was split into 3×3 square-foot cells (so $n = 48$). An indicator for army-worm damage was recorded for the centre square in each cell. When the grass was planted the plough and harrows ran down the columns.

| | Easting | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| Northing | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 |
| 1 | 1 | 1 | 1 | 0 | 0 | 0 | 0 | 1 |
| 2 | 1 | 1 | 1 | 0 | 0 | 1 | 0 | 1 |
| 3 | 0 | 1 | 1 | 1 | 1 | 0 | 1 | 1 |
| 4 | 1 | 1 | 1 | 0 | 0 | 1 | 1 | 1 |
| 5 | 1 | 1 | 0 | 0 | 0 | 0 | 0 | 1 |
| 6 | 0 | 0 | 1 | 1 | 0 | 1 | 1 | 1 |

For $i = 1, ..., n$, let $Y_i \in \{0, 1\}$ indicate army-worm damage observed in cell $i$. The data are displayed above. Suppose $Y_i \sim \text{Bernoulli}(\mu_i)$ independently in each cell, with

$$\mu_i = \frac{\exp(z_i)}{1 + \exp(z_i)}.$$

where $z_i \in \Re$ is an unobserved real latent variable associated with cell $i$. Let $z = (z_1, ..., z_n)$. The $z$-values are modelled by a conditional auto-regression with an $n \times n$ weight matrix $W$ and

$$z_i \mid z_{-i}, \sigma \sim N\left(\textstyle\sum_{j \neq i} W_{i,j} z_j, \sigma^2\right).$$

It is of interest to estimate a map of the log-odds for damage, and to test for an effect due to ploughing. Assume a prior for $\sigma$ is available and consider eliciting a prior for $z$ by specifying $W$.

(a) Suppose that for $i = 1, ..., n$, $W_{ii} = 0$ and $I - W$ is positive definite. Let $\pi(z|\sigma)$ denote the joint density of $z \in \mathbb{R}^n$ given $\sigma$. Show that

$$\pi(z|\sigma) = N(z; 0_n, \sigma^2(I - W)^{-1}).$$

**Solution:** We seek a multivariate normal joint density having the specified full conditional distributions. Suppose that

$$\pi(z \mid \sigma) \propto \exp\left\{-\frac{1}{2\sigma^2} z^{\mathsf{T}} Q z\right\},$$

where $Q$ is symmetric and positive definite. To find the conditional distribution of $z_i$, gather terms that depend on $z_i$:

$$z^{\mathsf{T}} Q z = Q_{ii} z_i^2 + 2 z_i \sum_{j \neq i} Q_{ij} z_j + h(z_{-i}),$$

where $h(z_{-i})$ does not depend on $z_i$. Completing the square gives

$$z^{\mathsf{T}} Q z = Q_{ii} \left( z_i + \frac{1}{Q_{ii}} \sum_{j \neq i} Q_{ij} z_j \right)^2 + \widetilde{h}(z_{-i}).$$

It follows that

$$z_i \mid z_{-i}, \sigma \sim N\left( -\frac{1}{Q_{ii}} \sum_{j \neq i} Q_{ij} z_j, \frac{\sigma^2}{Q_{ii}} \right).$$

Try $Q = I - W$. Since $W_{ii} = 0$, we have $Q_{ii} = 1$, while for $i \neq j$, $Q_{ij} = -W_{ij}$, so

$$z_i \mid z_{-i}, \sigma \sim N\left( \textstyle\sum_{j \neq i} W_{ij} z_j, \sigma^2 \right).$$

Finally, since $Q = I - W$ is symmetric and positive definite, the joint distribution is

$$z \mid \sigma \sim N\left(0_n, \sigma^2 Q^{-1}\right) = N\left\{0_n, \sigma^2 (I - W)^{-1}\right\}.$$

(b) Scientist $A$ believes a small number of army worm midges blew in after planting, so that the first arrivals were scattered at random over the area and these then expanded through the ground to form local clusters. Using the auto-regression, elicit a prior $\pi_A(z|\sigma, u)$ for $z$ with a single scalar parameter $u$.

**Solution:** Let $i \sim j$ if $i$ is a neighbor of $j$ on the lattice. The cells are all equally spaced and there is nothing to distinguish easting and northing. Also, we would expect some sort of local Markov structure, since the army-worm spread through the ground. We set $p(z_i|z_{-i}, \sigma) = p(z_i|z_{j\sim i}, \sigma, u)$ with $u > 0$ a coupling parameter appearing in $W^{(A)}$,

$$W_{i,j}^{(A)} = \begin{cases} u & \text{if } i \sim j \text{ and} \\ 0 & \text{otherwise.} \end{cases}$$

This gives

$$\pi_A(z|\sigma, u) = N(z; 0_n, \sigma^2(I - W^{(A)})^{-1}).$$

The matrix $W^{(A)}$ is symmetric because the neighbour relation is symmetric. Also, we must choose $u > 0$ sufficiently small that $I - W^{(A)}$ is positive definite.

(c) Scientist $B$ believes the army worms blew in before planting and formed small clusters a foot or so across which were subsequently smeared out along the columns by ploughing and seeding. Using the auto-regression, elicit a prior $\pi_B(z|\sigma, v)$ for $z$ with a single scalar parameter $v$.

**Solution:** By the same reasoning we set $p(z_i|z_{-i}, \sigma) = p(z_i|z_{j\sim i}, \sigma, v)$ with $v > 0$ a parameter in $W^{(B)}$,

$$W^{(B)}_{i,j} = \begin{cases} v & \text{if } i \sim j \text{ and } j \text{ is directly north or south of } i \\ 0 & \text{otherwise.} \end{cases}$$

This will make the columns independent in the prior. Note that $W^{(B)}$ is again symmetric and $v$ must be small enough as for $u$ above. This gives

$$\pi_B(z|\sigma, v) = N(z; 0_n, \sigma^2(I - W^{(B)})^{-1}).$$

(d) Suppose priors for $\sigma$, $u$ and $v$ have been elicited. Say how you would check that these priors and the priors for $z$ reflect available knowledge. Indicate any further information you would need from Scientists A and B.

**Solution:** We could simulate synthetic data by simulating $z' \sim \pi_A(z|\sigma, u)$ (or similarly B) and then for $i = 1, ..., n$ simulate $Y'_i \sim \text{Bernoulli}(\mu_i(z'_i))$. We could check with the scientists that $Y'$ shows plausible variation by estimating for example the correlation of $Y_i$ and $Y_j$ for $i$ and $j$ neighboring cells and $i$ and $j$ distant pairs. We would reference this to their previous experience with army-worm, not to the data in hand.

8. Consider a process generating $x_1, x_2, x_3, ...$ in which $x_1 = 1$ with fixed and known probability $0 < p < 1$ and for $n = 1, 2, ...,$

$$p(x_{n+1} = 1|x_n, ..., x_1) = \frac{p + k_n}{1 + n},$$

where $k_n = \sum_{i=1}^n x_i$ and $p(x_{n+1} = 0|x_n, ..., x_1) = 1 - p(x_{n+1} = 1|x_n, ..., x_1)$.

(a) Is the process Markov?

**Solution:** No. For eg $p(1|1,1) = (p+2)/3$ and $p(1|1,0) = (p+1)/3$ which are not equal for any $0 < p < 1$ so the process depends on its history.

(b) Show that this process generates an infinite exchangeable sequence.

**Solution:** If $q \sim \text{Beta}(\alpha, \beta)$ and $x_i = 1$ wp $q$ and otherwise $x_i = 0$ iid for $i = 1, ..., n$ then

$$p(x_1, ..., x_n|q) = q^k(1-q)^{n-k}$$

where $k = \sum_i x_i$. It follows that

$$\begin{aligned} p(x_1, ..., x_n) &= \int q^{k_n}(1-q)^{n-k_n} \frac{\Gamma(\alpha+\beta)}{\Gamma(\alpha)\Gamma(\beta)} q^{\alpha-1}(1-q)^{\beta-1} dq \\ &= \frac{\Gamma(\alpha+\beta)}{\Gamma(\alpha)\Gamma(\beta)} \frac{\Gamma(\alpha+k)\Gamma(\alpha+n-k)}{\Gamma(\alpha+\beta+n)} \end{aligned}$$

and so

$$\frac{p(x_1, ..., x_n, 1)}{p(x_1, ..., x_n)} = \frac{\alpha + k_n}{\alpha + \beta + n}.$$

Now $\Pr(x_1 = 1) = p$ so $p = E(E(x_1|q)) = \alpha/(\alpha+\beta)$ so if we take $\alpha = p$ and $\beta = 1-p$ then we match this and

$$p(x_{n+1} = 1|x_1, ..., x_n) = \frac{p + k_n}{1 + n}.$$

It follows by de Finetti that $x_1, x_2, ...$ is an infinite exchangeable sequence since

$$p(x_1, ..., x_n) = \int \prod_{i=1}^{n} q^{x_i}(1-q)^{1-x_i} \text{Beta}(q; p, 1-p) dq.$$

9. Let $G = (V, E)$ be a directed acyclic graph with nodes $V = \{1, ., ..., n\}$ and edges $E \subseteq V \times V$. A Directed Acyclic Graph (DAG) is a directed graph with no directed paths that form a cycle.

   For $n = 1, 2, 3, ...$ let $\mathcal{G}_n$ be the set of all distinct DAGs on $n$ labeled nodes and let $|\mathcal{G}_n|$ give the number of such DAGs. There are three DAGs on $n = 2$ nodes with $V = \{1, 2\}$ and $G_i = (V, E_i)$, $i = 1, 2, 3$. Possible edge sets are $E_1 = \emptyset$, $E_2 = \{(1, 2)\}$ and $E_3 = \{(2, 1)\}$, so $|\mathcal{G}_2| = 3$. There are $|\mathcal{G}_3| = 25$ DAGs on $n = 3$ nodes. See `https://oeis.org/A003024`.

   (a) Let $\pi_{U,n}(g)$, $g \in \mathcal{G}_n$ give the uniform distribution on DAGs with $n$ nodes, so that

   $$\pi_{U,n}(g) = \frac{\mathbb{I}_{g \in \mathcal{G}_n}}{|\mathcal{G}_n|}.$$

   For $G \sim \pi_{U,3}$ and $G = (V, E)$ let $G_{-3} \in \mathcal{G}_2$ be the DAG obtained from $G$ by removing node

3 and any edge $(i,3) \in E$ or $(3,i) \in E$, $i = 1,2$ connected to node 3. Show that $G_{-3}$ is not distributed according to $\pi_{U,2}$.

**Solution:** (3 marks, standard) 3 into 25 wont go. Let $g \in \mathcal{G}_2$ be the DAG $g = (V = \{1,2\}, E = \emptyset)$ so $\pi_{U,2}(g) = 1/3$. It is sufficient to show that

$$\pi_{U,2}(g) \neq \sum_{g' \in \mathcal{G}_3} \pi_{U,3}(g') \mathbb{I}_{g'_{-3}=g}.$$

The LHS is equal 1/3. The set $\{g' \in \mathcal{G}_3 : g'_{-3} = g\}$ has 9 elements (one with 0 edges, 4 with one edge and 4 with two edges) so since $1/3 \neq 9/25$ we have $\Pr(G_{-3} = g) \neq \pi_{U,2}(g)$. We dont actually have to count DAGs in $\{g' \in \mathcal{G}_3 : g'_{-3} = g\}$ as there is no integer $m$ such that $m/25 = 1/3$.

(b) (research question) Construct a family of probability distributions $p_{1:n}(g),\ g \in \mathcal{G}_n$ over DAGs that is marginally consistent, in the sense that if $G \sim p_{1:n}$ then $G_{-n} \sim p_{1:n-1}$. Of course we can "cheat" and write down any $p_{1:n}$ we like for some fixed $n$ and then *define* the probability $p_s,\ s \subset [n]$ for DAGs on the vertex set $s$ to be the marginal! Not very practical at large $n$.

10. Let $h$ be a simple directed graph (allow cycles) so for $(i,j) \in I_n$, with

$$I_n = \{(i,j) \in [n] \times [n] : i < j\}$$

the possible values are $h_{i,j} \in \Omega_{i,j}$, where

$$\Omega_{i,j} = \{i \rightarrow j, i \leftarrow j, i \sim j\}$$

and $i \sim j$ means that there is no edge between $i$ and $j$. Let $\mathcal{H}_n$ be the space of possible simple directed graphs on $n$ vertices. Let $H \in \mathcal{H}_n$ be the unknown true graph and suppose that we have data $y$ informing $H$, a prior $\pi(h)$, and a posterior $\pi(h \mid y)$, $h \in \mathcal{H}_n$. Let $\hat{h} \in \mathcal{H}_n$ be an estimate of $H$ and let

$$l(\hat{h}, H) = n(n-1)/2 - \sum_{i<j} \mathbb{I}_{\hat{h}_{i,j}=H_{i,j}}$$

be the loss if we estimate $\hat{h}$ when the truth is $H$.

(a) For $(i,j) \in I_n$ and $e \in \Omega_{i,j}$, let $p(e) = P(H_{i,j} = e \mid y)$ be the marginal posterior probability that $H_{i,j} = e$. Show that the Bayes estimator

$$\hat{h} = \arg\min_{h \in \mathcal{H}_n} E_{H|y}\{l(h,H)\}$$

is given by $\hat{h}_{i,j} = \arg\max_{e \in \Omega_{i,j}} p(e)$.

**Solution:** For $h \in \mathcal{H}_n$, let

$$\vec{S}(h) = \{(i,j) \in I_n : h_{i,j} = i \to j\},$$

and let $\overleftarrow{S}(h)$ and $\widetilde{S}(h)$ be defined similarly. The Expected Posterior Loss (EPL) is

$$\begin{aligned} E_{H|y}\{l(h,H)\} &= \frac{n(n-1)}{2} - \sum_{i<j} E_{H|y}\left(\mathbb{I}_{h_{i,j}=H_{i,j}}\right) \\ &= \frac{n(n-1)}{2} - \sum_{i<j} P(H_{i,j} = h_{i,j} \mid y) \\ &= \frac{n(n-1)}{2} - \sum_{(i,j)\in\vec{S}(h)} p(i\to j) - \sum_{(i,j)\in\overleftarrow{S}(h)} p(i\leftarrow j) - \sum_{(i,j)\in\widetilde{S}(h)} p(i \sim j). \end{aligned}$$

Thus, we choose $h$ to make the three sums as large as possible. Since

$$\left(\vec{S}(h), \overleftarrow{S}(h), \widetilde{S}(h)\right)$$

is a partition of $I_n$, each $(i,j)$ appears in exactly one of the sums. We should therefore choose each $h_{i,j}$ so that $p(h_{i,j})$ is as large as possible. Hence

$$\hat{h}_{i,j} = \arg\max_{e\in\Omega_{i,j}} p(e).$$

(b) Rewrite the loss in terms of reward and utility, and give the posterior reward distribution in terms of $\pi(\cdot \mid y)$. Relate the expected utility to the Expected Posterior Loss and show that the Bayes estimator maximises the expected utility.

**Solution:** Fix $h \in \mathcal{H}_n$. If $H \sim \pi(\cdot \mid y)$, then the reward

$$R(h,H) = \sum_{i<j} \mathbb{I}_{h_{i,j}=H_{i,j}}$$

takes values in

$$\mathcal{R}_n = \left\{0, 1, \ldots, \frac{n(n-1)}{2}\right\}.$$

If we take the utility to be $U(R) = R$, then

$$l(h,H) = \frac{n(n-1)}{2} - U\{R(h,H)\}.$$

Partition $\mathcal{H}_n$ into the sets

$$\mathcal{H}_n = \bigcup_{r\in\mathcal{R}_n} \mathcal{H}_{n,r}, \qquad \mathcal{H}_{n,r} = \{h' \in \mathcal{H}_n : R(h,h') = r\},$$

whose elements give equal reward. The posterior reward distribution

$$P_{h,y}(r) = P\{R(h,H) = r \mid y\}$$

is

$$P_{h,y}(r) = \sum_{h' \in \mathcal{H}_{n,r}} \pi(h' \mid y), \qquad r \in \mathcal{R}_n.$$

Consequently,

$$\begin{aligned}
E_{H|y}\{l(h,H)\} &= \frac{n(n-1)}{2} - \sum_{h' \in \mathcal{H}_n} \pi(h' \mid y) U\{R(h,h')\} \\
&= \frac{n(n-1)}{2} - \sum_{r \in \mathcal{R}_n} \left[ \sum_{h' \in \mathcal{H}_{n,r}} \pi(h' \mid y) \right] U(r) \\
&= \frac{n(n-1)}{2} - \sum_{r \in \mathcal{R}_n} P_{h,y}(r) U(r) \\
&= \frac{n(n-1)}{2} - E_{P_{h,y}}\{U(R)\},
\end{aligned}$$

so we minimise the EPL $E_{H|y}\{l(h,H)\}$ by maximising the expected utility $E_{P_{h,y}}\{U(R)\}$.

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